

IE3101 – Enterprise Network and Information Systems Security
Group Presentation
Client: Prismo Constructions (PVT) Ltd
Project: ConstructFlow – Secure Network Design (Option B)

Slide 1: Title Slide

Network Security Assessment and Secure Network Design

Client: Prismo Constructions (PVT) Ltd

System: ConstructFlow – Digital Project Management Platform

Option B – New / Expanded Network

Group Members:

- Member 1
- Member 2
- Member 3
- Member 4

Module: IE3101 – Enterprise Network and Information Systems Security
Year 3 Semester 1 | 2026

Slide 2: Agenda

1. Client Context & Option Selection
 2. Current Baseline Network
 3. Network Requirements & Critical Assets
 4. Vulnerability Identification
 5. Risk Assessment & Prioritization
 6. ISO/IEC 27001 Alignment & Toolkit Application
 7. To-Be Secure Network Architecture
 8. Prioritized Recommendations
 9. Conclusion & Key Takeaways
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Slide 3: Client Context

Prismo Constructions (PVT) Ltd

- Construction consultancy company in Sri Lanka
- Expanding from consultancy into **construction management**
- Currently operates with limited IT infrastructure

- Heavy reliance on email, WhatsApp, and manual processes
- Introducing **ConstructFlow** – a web-based project management platform (React + Spring Boot)

Key Users:

- CEO
 - Project Manager
 - Site Engineers
 - Administrative Officer
 - External Clients
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Slide 4: Why Option B?

Selected Option: B – New / Expanded Network

Justification:

- Current network is very limited (basic internet + individual laptops)
 - No centralized servers, no structured remote access, no segmentation
 - Organization is expanding operations and introducing a new critical system
 - Need to design a secure network **before** full implementation
 - Supports security-by-design approach
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Slide 5: Current Baseline Network (As-Is)

Current Environment Characteristics:

- Individual laptops connected to basic internet
- Marketing website only
- No internal servers for applications or databases
- No firewall or network segmentation
- Communication via email and WhatsApp
- Progress photos and documents shared informally
- Weak or no centralized authentication

Major Limitation:

Unable to securely support multi-user project management with remote Site Engineers and external Clients.

Slide 6: Network Requirements

Key Requirements for ConstructFlow:

- Secure access for Office staff, Site Engineers, and Clients
 - Centralized storage of project data, progress logs, and photographs
 - Role-based access control
 - Encrypted remote access
 - Protection of confidential project and client information
 - Reasonable availability during working hours
 - Scalable for future growth
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Slide 7: Critical Assets

Most Important Assets:

Asset	Sensitivity
ConstructFlow Application	High
Project Database	High
Progress Photographs	High
Client Documents & Contracts	High
User Credentials	High
Approval Records	High
Office & Field Laptops	Medium-High

Slide 8: Key Vulnerabilities Identified

Major Vulnerabilities (Client-Specific):

- Weak authentication and absence of MFA
- No Role-Based Access Control
- Insecure sharing of progress photos and documents (WhatsApp/Email)
- Weak remote access controls
- Lack of network segmentation and firewall
- Limited endpoint protection on field devices

- No centralized logging or monitoring
 - Inadequate backup procedures
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Slide 9: Risk Assessment Summary

Top Critical / High Risks:

Risk ID	Risk Description	Level
R01	Weak authentication / No MFA	Critical
R02	Insufficient Role-Based Access Control	Critical
R03	Insecure document & photo handling	Critical
R04	Weak remote access controls	High
R05	Limited firewall & no segmentation	High
R06	Lost/stolen devices with weak protection	High

Slide 10: ISO/IEC 27001 Toolkit Application

Toolkit Materials Used:

- Asset Register
- Risk Register
- Statement of Applicability (SoA)
- Information Classification Matrix
- RASCI Table
- ISMS & Controls Status (Gap Analysis)

Approach:

Applied toolkit to the **proposed** network environment (Option B) to support security-by-design.

Slide 11: Key Selected Controls (SoA)

Most Important Applicable Controls:

- Access Control Policy & RBAC (A.9)
- Secure authentication & MFA (A.9.4)
- Cryptographic controls – HTTPS & encryption (A.10)

- Network segregation (A.13.1.3)
 - Information backup (A.12.3)
 - Event logging (A.12.4)
 - Security awareness (A.7.2.2)
 - Information transfer controls (A.13.2)
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Slide 12: To-Be Secure Network Architecture

High-Level Design Principles:

- Defense in Depth
- Least Privilege
- Network Segmentation
- Secure Remote Access
- Practicality for a small organization

Main Zones:

- Office User Zone
 - Server / Application Zone (Highest trust)
 - Remote Access Zone (Site Engineers)
 - External / Client Access Path
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Slide 13: To-Be Network – Key Features

Core Security Features:

- Firewall as primary trust boundary
 - ConstructFlow Application + Database in protected Server Zone
 - Encrypted access (HTTPS) for all users
 - Role-Based Access Control
 - Multi-Factor Authentication (planned)
 - Centralized secure storage for photos and documents
 - Basic logging and backup
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Slide 14: Prioritized Recommendations

Implementation Priority:

Immediate (0–1 Month)

- Strong password policy + MFA
- Role-Based Access Control
- HTTPS only
- Move documents & photos into ConstructFlow

Short-Term (1–3 Months)

- Firewall deployment
- Endpoint protection
- Basic logging
- Security awareness briefing

Medium-Term (3–6 Months)

- Formal backup & recovery
- Access rights review
- Basic security policies

Slide 15: Expected Risk Reduction

After implementing the recommended controls:

- Most Critical and High risks expected to reduce to **Low** or **Low-Medium** residual risk
- Significant improvement in Confidentiality, Integrity, and Availability
- Secure foundation for ConstructFlow operations

Slide 16: Conclusion

Key Outcomes:

- Current environment is insufficient for secure construction project management
- Clear set of client-specific vulnerabilities and risks identified
- ISO 27001 Toolkit effectively applied for risk treatment
- Practical and scalable To-Be Secure Network Architecture designed
- Prioritized, realistic recommendations provided

Final Message:

Prismo Constructions can securely expand into construction management by implementing the recommended controls in a phased and practical manner.

Slide 17: Thank You

Thank You

Questions & Discussion

Group Members ready for Viva

Notes for your team:

- Total slides: 17 (ideal length for the presentation)
- Keep each slide clean (maximum 5–6 bullet points)
- Use the diagrams we prepared earlier (As-Is in early slides, full To-Be in slides 12–13)
- Practice explaining the difference between Task 2 (high-level) and Task 6 (detailed To-Be)